POWER BI PROJECT REPORT

Movie Analytics and Insights

An interactive 3-page Power BI dashboard on movies, genres, ratings, popularity and release trends

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PROJECT TITLE	Movie Analytics and Insights
TOOL USED	Microsoft Power BI Desktop
PROJECT TYPE	Business Intelligence / Data Visualization

9,817	13.7M	6.4	5.1K
Total Movies	Total Votes	Average Rating	Top Popularity

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ABOUT THIS REPORT

- An interactive 3-page Power BI dashboard that analyses 9,817 movies by genre, rating, popularity, votes and release year.
- Dashboard values quoted in this report are read from the Power BI screenshots in Section 7. The full project (Power BI file, dataset, DAX code and README) is available in the GitHub repository.

1 Student and Project Details

Student Name	Pratham Deep
Registration Number	PIET24CD063
Batch / Roll No.	G3 / 063
Project Title	Movie Analytics and Insights
Tool Used	Microsoft Power BI Desktop (Power Query, Data Modeling, DAX)
Project Type	Business Intelligence / Data Visualization

Movie Analytics and Insights is an end-to-end Power BI project that turns a raw movie dataset into an interactive dashboard. It analyses movies by genre, rating, popularity, votes and release year, and lets the viewer filter every visual with seven slicers, including a movie-title search box.

9,817	13.7M	6.44	5.1K
Total Movies	Total Votes	Avg Rating (gauge)	Top Popularity

Dashboard page	Purpose	Main visuals
Movie Details	Overview and drill-down by movie	KPI cards, rating gauge, movie table, Popularity vs Rating scatter, Movies by Release Year, Top Rated Movies
Movies Insights	Popularity behaviour	Movies Distribution by popularity tier, monthly popularity trend, Movies by Release Year (donut)
Movie Info	Ratings, genres and decades	Movie Ratings pie, Movie Trends by Decade and Genre, Movies by Primary Genre

2 Dataset Description

The dataset (**mymoviedb.csv**) holds **9,827 movie records** with 9 columns. It covers movies released from **1902 to 2024** in **43 original languages** (English is about 77%) across **19 genres**. After preparation, the Power BI model **mymoviedb** contains **9,817 movies**. Every movie has a popularity score, a vote count and an average rating from 0 to 10; 100 movies have no votes yet.

Column	Type	Description
Release_Date	Date	Release date of the movie
Title	Text	Movie title
Overview	Text	Short plot summary
Popularity	Decimal	Popularity score (maximum 5,083.954 = 5.1K)
Vote_Count	Whole number	Number of votes received
Vote_Average	Decimal	Average rating on a 0-10 scale
Original_Language	Text	Original language code (en, ja, es, fr, ko ...)
Genre	Text	All genres of the movie, comma separated
Poster_Url	Text	Poster link (not used in the model)

Columns added during preparation: **Genre Primary**, **Release Year**, **Release Decade**, **Rating_Category**, **Popularity_Tier**, their sort-order helper columns and **Certification**.

3 Objectives

- Build one overview of total movies, total votes, average rating and top popularity.
- Understand how movie releases have grown over the years and decades.
- Classify movies by rating quality (Poor to Excellent) and by popularity tier (Low to Very High).
- Identify the top-rated movies and the most common genres.
- Study the relationship between popularity and rating.
- Provide interactive exploration with slicers, a movie-title search box and drill-through.
- Convert raw movie data into clear, actionable insights.

4 Data Cleaning Performed

Data preparation was done in the Power Query Editor before modelling. The steps are also provided as M code in the GitHub repository (`DAX/power_query_steps.m`).

9,827	10	9,817	100
Records in source CSV	Rows excluded	Movies in model	Kept as Not Rated

Step	Action	Why
1	Loaded the CSV and promoted the first row to headers	Get clean column names
2	Set data types: date, whole number, decimal, text	Correct sorting, maths and date hierarchy
3	Removed broken and blank rows. One record (Pixie Hollow Bake Off) has an overview that wraps over 11 lines and loads as null fragments	Avoid null / garbage rows in counts and averages
4	Dropped the Poster_Url column	Not used in any visual
5	Created Genre Primary (first listed genre) and Release Year	One genre per movie for slicers and genre charts; year for trends
6	Created Release Decade, Rating_Category and Popularity_Tier, with sort-order helper columns	Group movies into meaningful, correctly ordered buckets
7	Kept the 100 movies with 0 votes and labelled them Not Rated	Do not lose movies; do not let 0 ratings distort categories
8	Kept titles with special characters such as #Alive, [REC], *batteries not included	They are genuine movie titles

CLASSIFICATION RULES

- Rating_Category: Not Rated = 0 votes; Poor below 5; Below Average 5 to 6; Average 6 to 7; Good 7 to 8; Excellent 8 and above.
- Popularity_Tier: Low below 40; Moderate 40 to 100; High 100 to 300; Very High 300 and above.

5 Data Model

The model is kept simple: one flat fact table (**mymoviedb**) holds every field used by the visuals, and a dedicated measures table (**_Measures**) keeps all DAX measures in one place. The measures read the fact table directly, so no physical relationship is needed. All slicers filter every visual on the page together, and drill-through is enabled with Cross-report and Keep all filters turned on.

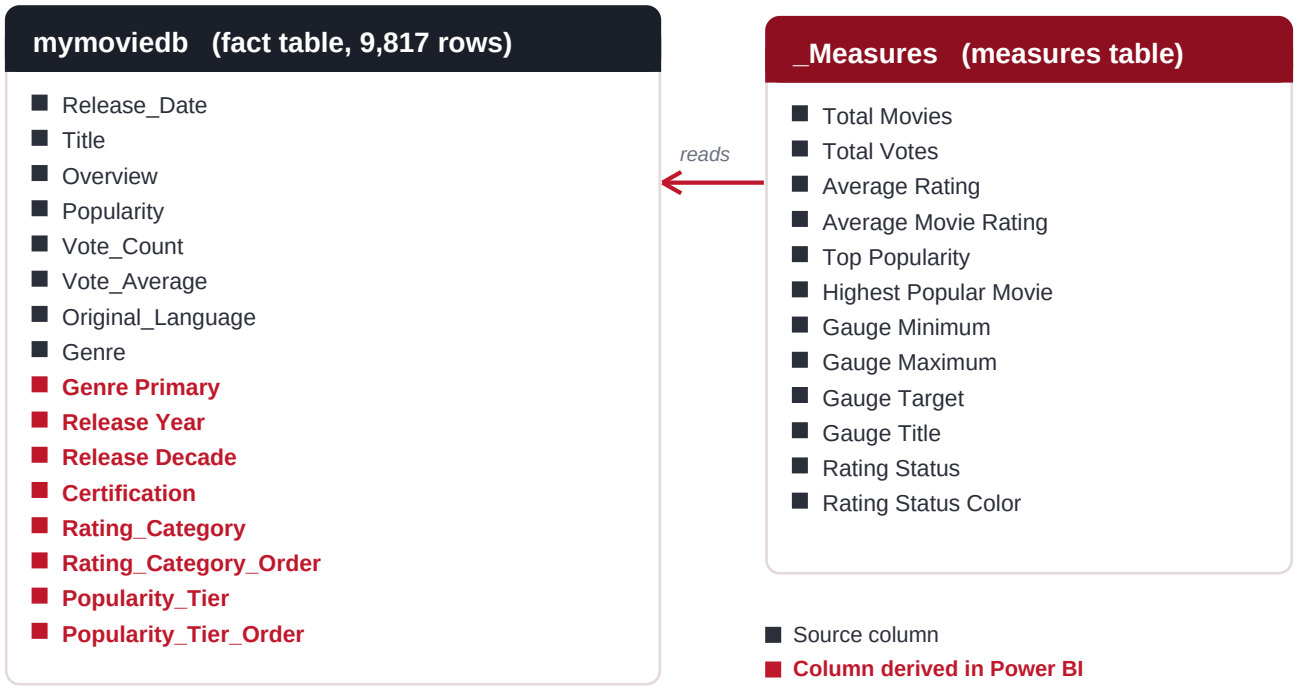


Figure 1: Data model - fact table and measures table

Power BI Data view of the model

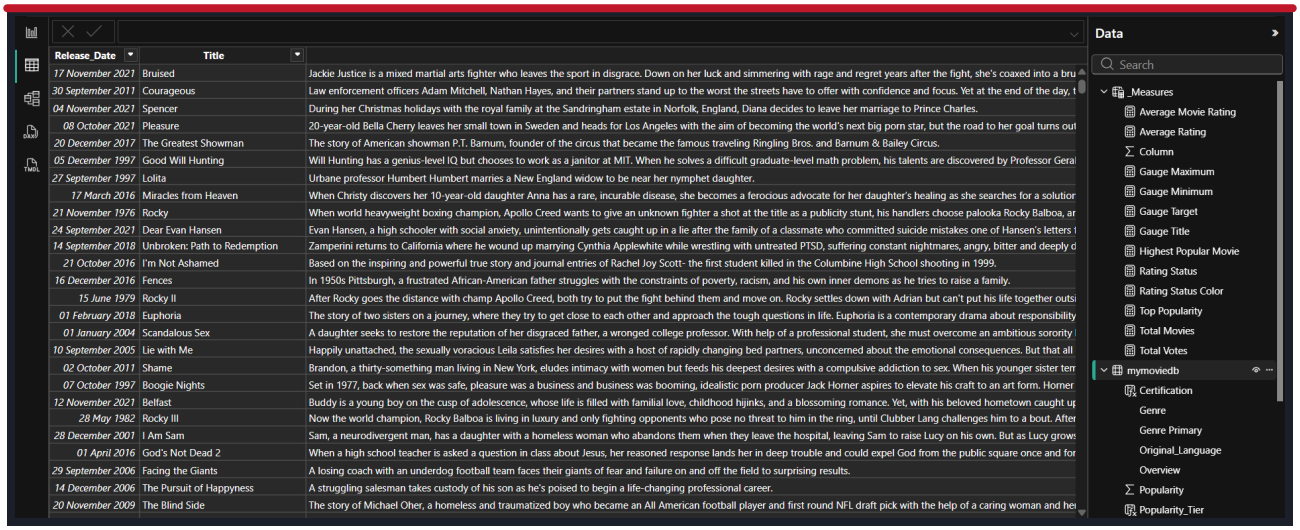
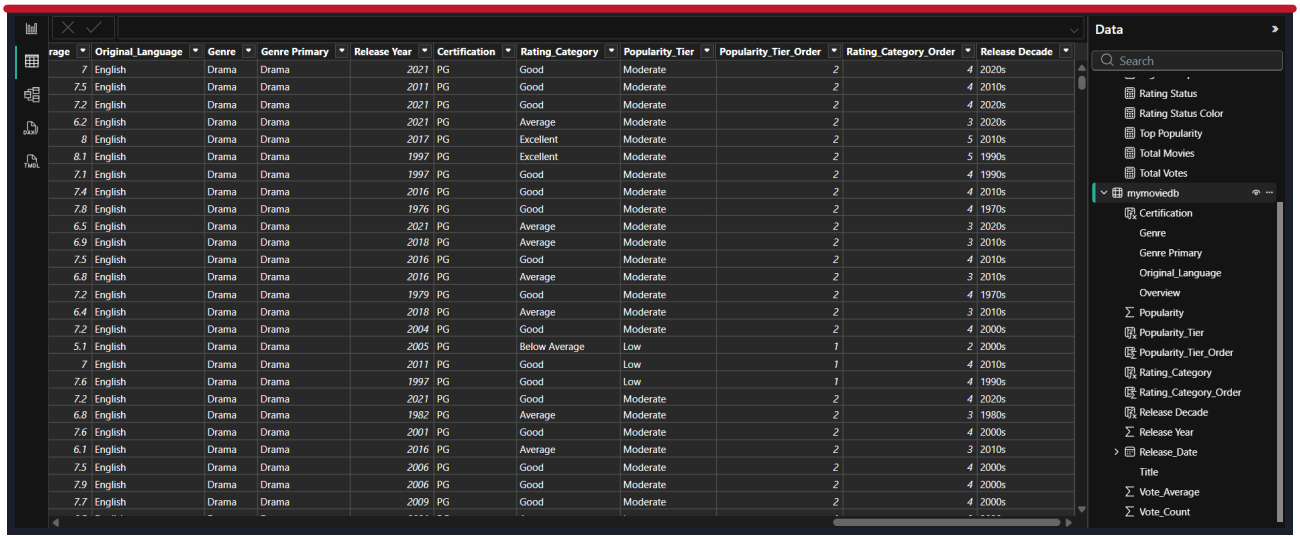


Figure 2: Data view - mymoviedb table (left) and the _Measures table (right)



rage	Original_Language	Genre	Genre Primary	Release Year	Certification	Rating_Category	Popularity_Tier	Popularity_Tier_Order	Rating_Category_Order	Release Decade
7	English	Drama	Drama	2021	PG	Good	Moderate	2	4	2020s
7.5	English	Drama	Drama	2011	PG	Good	Moderate	2	4	2010s
7.2	English	Drama	Drama	2021	PG	Good	Moderate	2	4	2020s
6.2	English	Drama	Drama	2021	PG	Average	Moderate	2	3	2020s
8	English	Drama	Drama	2017	PG	Excellent	Moderate	2	5	2010s
8.1	English	Drama	Drama	1997	PG	Excellent	Moderate	2	5	1990s
7.1	English	Drama	Drama	1997	PG	Good	Moderate	2	4	1990s
7.4	English	Drama	Drama	2016	PG	Good	Moderate	2	4	2010s
7.8	English	Drama	Drama	1976	PG	Good	Moderate	2	4	1970s
6.5	English	Drama	Drama	2021	PG	Average	Moderate	2	3	2020s
6.9	English	Drama	Drama	2018	PG	Average	Moderate	2	3	2010s
7.5	English	Drama	Drama	2016	PG	Good	Moderate	2	4	2010s
6.8	English	Drama	Drama	2016	PG	Average	Moderate	2	3	2010s
7.2	English	Drama	Drama	1979	PG	Good	Moderate	2	4	1970s
6.4	English	Drama	Drama	2018	PG	Average	Moderate	2	3	2010s
7.2	English	Drama	Drama	2004	PG	Good	Moderate	2	4	2000s
5.1	English	Drama	Drama	2005	PG	Below Average	Low	1	2	2000s
7	English	Drama	Drama	2011	PG	Good	Low	1	4	2010s
7.6	English	Drama	Drama	1997	PG	Good	Low	1	4	1990s
7.2	English	Drama	Drama	2021	PG	Good	Moderate	2	4	2020s
6.8	English	Drama	Drama	1982	PG	Average	Moderate	2	3	1980s
7.6	English	Drama	Drama	2001	PG	Good	Moderate	2	4	2000s
6.1	English	Drama	Drama	2016	PG	Average	Moderate	2	3	2010s
7.5	English	Drama	Drama	2006	PG	Good	Moderate	2	4	2000s
7.9	English	Drama	Drama	2006	PG	Good	Moderate	2	4	2000s
7.7	English	Drama	Drama	2009	PG	Good	Moderate	2	4	2000s

Figure 3: Data view - derived columns Genre Primary, Release Year, Rating_Category, Popularity_Tier and Release Decade

6 DAX Measures

All measures live in the **_Measures** table; classification columns live in **mymoviedb**. The complete, commented code is in `DAX/dax_measures.dax` in the GitHub repository.

KPI measures

```
Total Movies          = COUNTROWS ( mymoviedb )                -- 9,817
Total Votes           = SUM ( mymoviedb[Vote_Count] )          -- 13.7M
Average Rating        = AVERAGE ( mymoviedb[Vote_Average] )    -- 6.4
Average Movie Rating  = AVERAGE ( mymoviedb[Vote_Average] )    -- 6.44
Top Popularity        = MAX ( mymoviedb[Popularity] )          -- 5.1K
```

Highest Popular Movie

```
Highest Popular Movie =
VAR _TopMovie =
    TOPN (
        1,
        VALUES ( mymoviedb[Title] ),
        CALCULATE ( MAX ( mymoviedb[Popularity] ) ),
        DESC
    )
RETURN
    MAXX ( _TopMovie, mymoviedb[Title] )
```

Gauge and rating status measures

```
Gauge Minimum = 0
Gauge Maximum = 10
Gauge Target  = 8
Gauge Title   = "Rating - Average"

Rating Status =
VAR _Rating = [Average Movie Rating]
RETURN
    SWITCH (
        TRUE (),
        _Rating >= 8, "Excellent",
        _Rating >= 7, "Good",
        _Rating >= 6, "Average",
        _Rating >= 5, "Below Average",
        "Poor"
    )
```

Calculated columns (table: mymoviedb)

```
Release Decade =
( mymoviedb[Release Year] - MOD ( mymoviedb[Release Year], 10 ) ) & "s"

Rating_Category =
IF (
    mymoviedb[Vote_Count] = 0,
    "Not Rated",
    SWITCH (
        TRUE (),
        mymoviedb[Vote_Average] >= 8, "Excellent",
        mymoviedb[Vote_Average] >= 7, "Good",
        mymoviedb[Vote_Average] >= 6, "Average",
        mymoviedb[Vote_Average] >= 5, "Below Average",
        "Poor"
    )
)

Popularity_Tier =
SWITCH (
    TRUE (),
    mymoviedb[Popularity] >= 300, "Very High",
    mymoviedb[Popularity] >= 100, "High",
    mymoviedb[Popularity] >= 40, "Moderate",
    "Low"
)
```

Item	Type	Used for
Rating_Category_Order, Popularity_Tier_Order	Calculated columns	Sort-by-column so categories and tiers appear in logical order (Poor to Excellent, Low to Very High)
Rating Status Color	Measure	Colour of the rating status text, based on the average rating

7 Dashboard Screenshots

The report has three pages that share one dark theme with red accents. Slicers (Genre(s), Title search, Certification, Release Year, Language, Primary Genre and Ratings) and the rating gauge are on every page, with KPI cards on the left edge. Each page is followed by the main insights it reveals.

Page 1 - Movie Details

Slicers, rating gauge, movie table, Popularity vs Rating scatter plot, KPI cards, Movies by Release Year and Top Rated Movies.

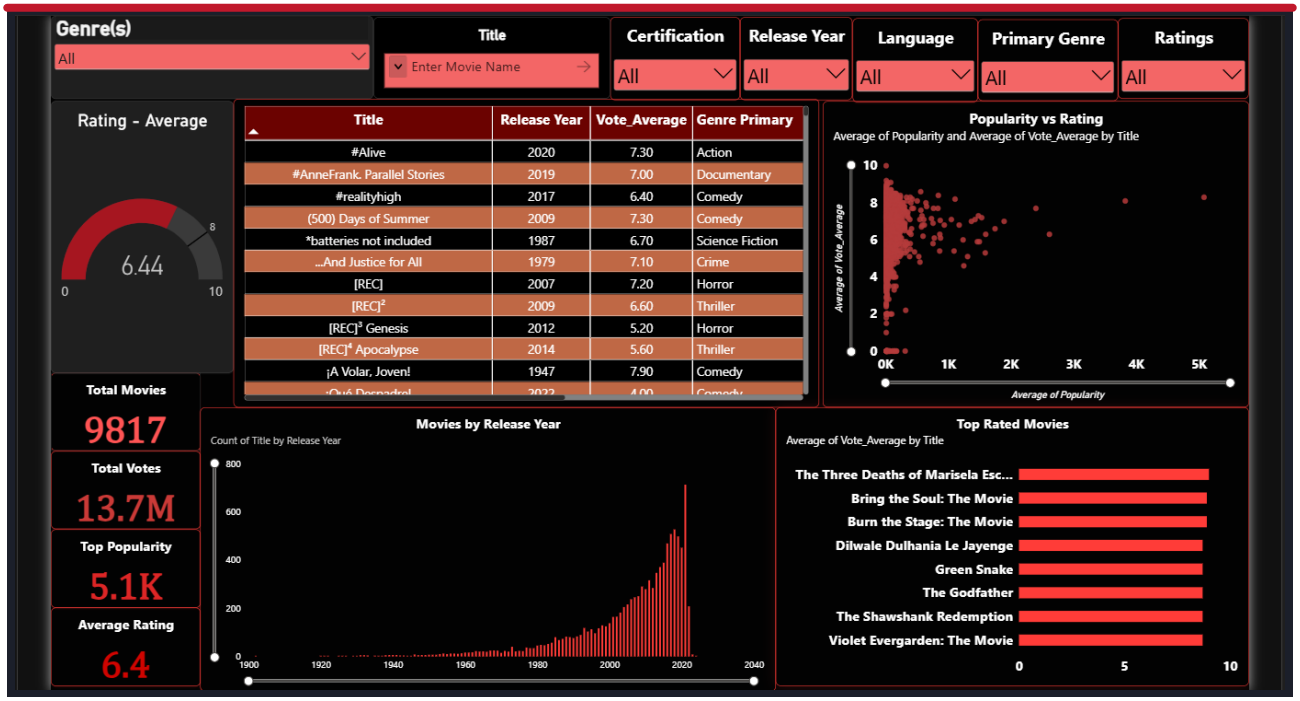


Figure 4: Page 1 - Movie Details

KEY INSIGHTS

- 9,817 movies with 13.7M total votes; average rating 6.4 (gauge 6.44 out of 10, target 8) and top popularity 5.1K.
- Releases climb steeply from the 1980s and peak at just over 700 movies in 2021.
- Most movies have popularity below 1K; only a few outliers reach 2K to 5K. The highest ratings (around 9) belong to low-popularity movies.
- Top rated movies include The Godfather, The Shawshank Redemption, Dilwale Dulhania Le Jayenge, Bring the Soul: The Movie and Burn the Stage: The Movie.

Page 2 - Movies Insights

Movies Distribution by popularity tier, monthly popularity trend with the highest popular movie, and Movies by Release Year.

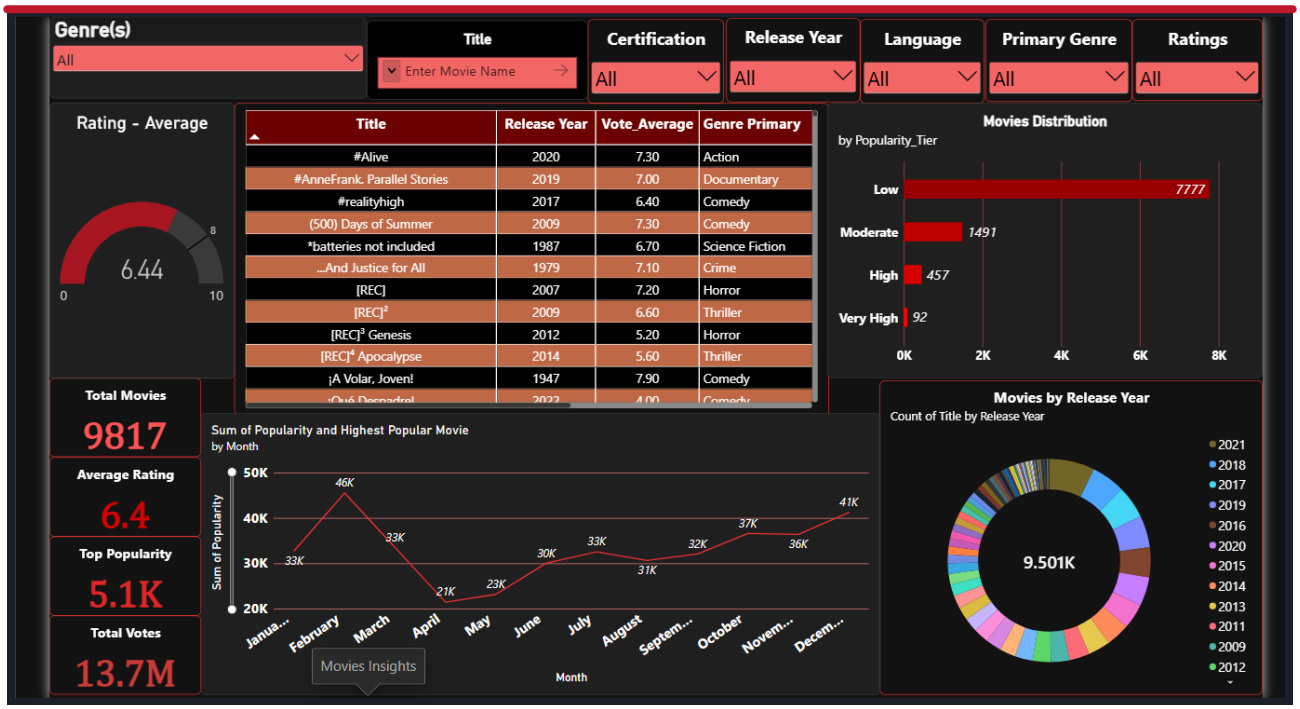


Figure 5: Page 2 - Movies Insights

KEY INSIGHTS

- Low-popularity movies dominate: 7,777 Low, 1,491 Moderate, 457 High and only 92 Very High.
- Total popularity by month peaks in February (46K), dips to its lowest in April (21K) and climbs back to 41K in December.
- The donut chart shows 2021, 2018, 2017, 2019 and 2016 as the release years with the most movies.

Page 3 - Movie Info

Movie Ratings by category, Movie Trends by Decade and Genre, and Movies by Primary Genre.

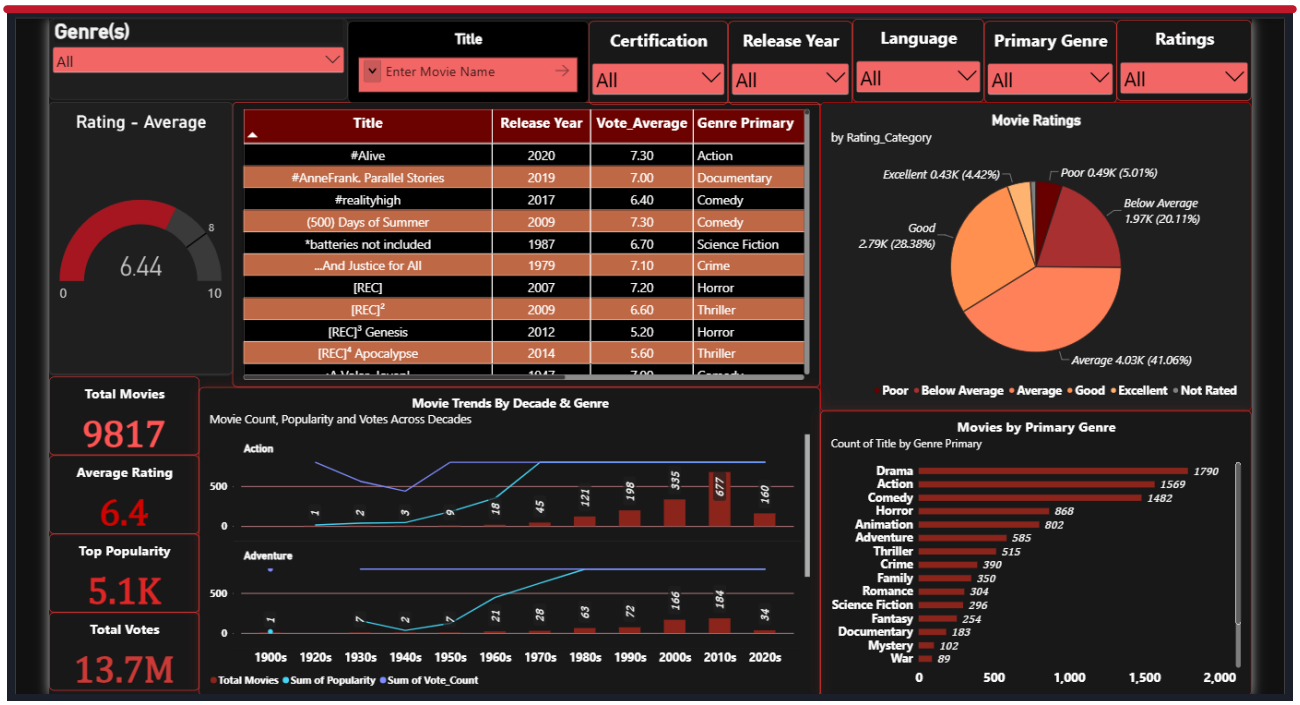


Figure 6: Page 3 - Movie Info

KEY INSIGHTS

- Average-rated movies are the largest group (4.03K, 41.06%), followed by Good (2.79K, 28.38%) and Below Average (1.97K, 20.11%).
- Poor (0.49K, 5.01%) and Excellent (0.43K, 4.42%) are the smallest rated groups.
- Drama (1,790), Action (1,569) and Comedy (1,482) are the most common primary genres, followed by Horror (868) and Animation (802).
- Action movies grew from 121 in the 1980s to 677 in the 2010s; Adventure grew from 63 to 184 over the same period.

8 Key Findings

9,817	41%	79%	49%
Movies analysed	Movies rated Average	Low popularity	Drama + Action + Comedy

Movies by rating category (thousands)

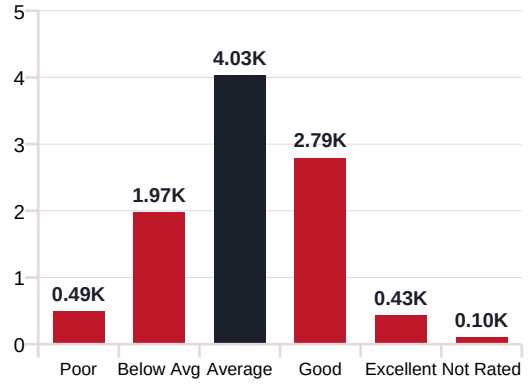


Figure 7: Rating categories (Movie Info page; Not Rated is about 0.10K)

Movies by popularity tier

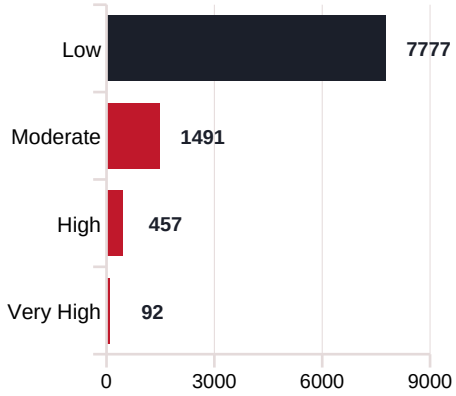


Figure 8: Popularity tiers (Movies Insights page)

Top 10 primary genres (movies)

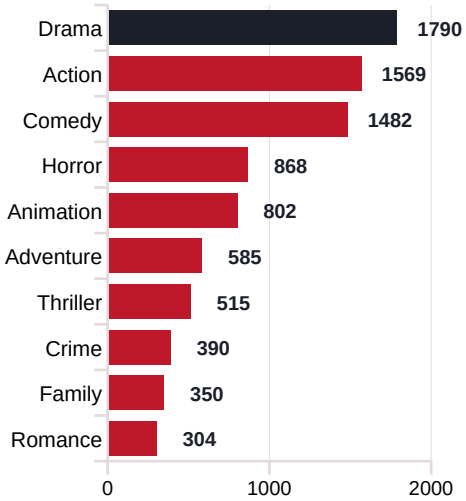


Figure 9: Top 10 primary genres (Movie Info page)

Total popularity by month (thousands)

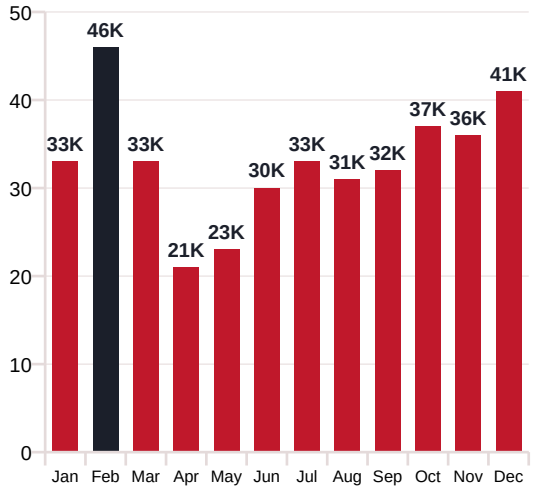


Figure 10: Monthly popularity (Movies Insights page)

Catalogue overview

- The catalogue holds 9,817 movies with 13.7M votes. The average rating is 6.44 out of 10, below the gauge target of 8.
- The highest popularity score is 5.1K.

Ratings and popularity

- Average (41.06%) and Good (28.38%) make up about 69% of movies; only 4.42% are Excellent and 5.01% are Poor.

- About 79% of movies are in the Low popularity tier; just 92 movies (under 1%) are Very High.
- Popularity does not guarantee quality: the highest ratings (around 9) belong to low-popularity movies, while the few very popular movies (2K to 5K) are rated about 6 to 8.5.

Genres, time and top movies

- Drama, Action and Comedy together account for about 49% of all movies.
- Production grew sharply: over 700 movies were released in 2021, and Action releases rose from 121 (1980s) to 677 (2010s).
- Monthly popularity is highest in February (46K), lowest in April (21K) and rises to 41K by December.
- Top rated titles include The Godfather, The Shawshank Redemption, Dilwale Dulhania Le Jayenge, Green Snake, Bring the Soul: The Movie and Burn the Stage: The Movie.

9 Conclusion

This project shows how Power BI can turn a raw movie dataset into an interactive analysis tool. The three-page dashboard covers headline KPIs, release and popularity trends, rating quality, genre mix and top-rated films, with seven slicers and a title search box for flexible exploration. The work covered the full BI workflow: cleaning in Power Query, a simple fact-plus-measures data model, DAX measures and calculated columns, and visual storytelling.

The analysis shows that the catalogue is dominated by average-quality, low-popularity titles, that Drama, Action and Comedy lead the genre mix, and that movie production grew sharply in recent decades. A small number of highly popular and highly rated movies stand out against this long tail, and popularity alone is not a sign of quality.

FUTURE IMPROVEMENTS

- Add budget and revenue data to measure profitability by genre and year.
- Add forecasting for release counts and popularity.
- Add language-wise and certification-wise analysis pages.
- Publish to Power BI Service with scheduled refresh and row-level security.

GitHub repository contents

Folder / file	Contents
README.md	Project overview, findings and how to open the project
Dataset/	dataset.xlsx (cleaned data, data dictionary, summary) and mymoviedb.csv (raw)
PowerBI/	Movie_analytics.pbix
DAX/	dax_measures.dax and power_query_steps.m
Screenshots/	Dashboard pages and Power BI data view screenshots
Report/	This project report (PDF)